Latent Semantic Analysis

Boston University CS 506 - Lance Galletti

Warm Up: working with text data

Warm Up: working with text data

Paris

Warm Up: working with text data

Paris France

Warm Up: working with text data

Paris France Rome

Warm Up: working with text data

Paris France Rome Italy

Warm Up: working with text data

Paris France Rome Italy

How do we convert these words to numbers that can be used by algorithms?

Warm Up: working with text data

Paris	France	Rome	Italy
1	2	3	4

Warm Up: working with text data

Paris	France	Rome	Italy
1	2	3	4

Paris and France close ✅
Rome and Italy close ✅
France and Rome equally close to Paris and France ❌

Warm Up: working with text data

France	Paris	Rome	Italy
1	2	3	4

Paris and France close ✅
Rome and Italy close ✅
France and Italy twice as far as Paris and Rome ❌

Warm Up: working with text data

Paris Rome

France Italy

Warm Up: working with text data

Paris Rome

France Italy

Paris and France close ✅
Rome and Italy close ✅
France and Italy twice as far as Paris and Rome ✅
Paris and Italy as far as Rome and France ✅

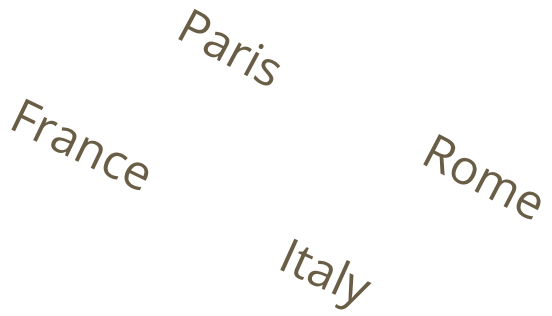
Warm Up: working with text data

Paris Rome

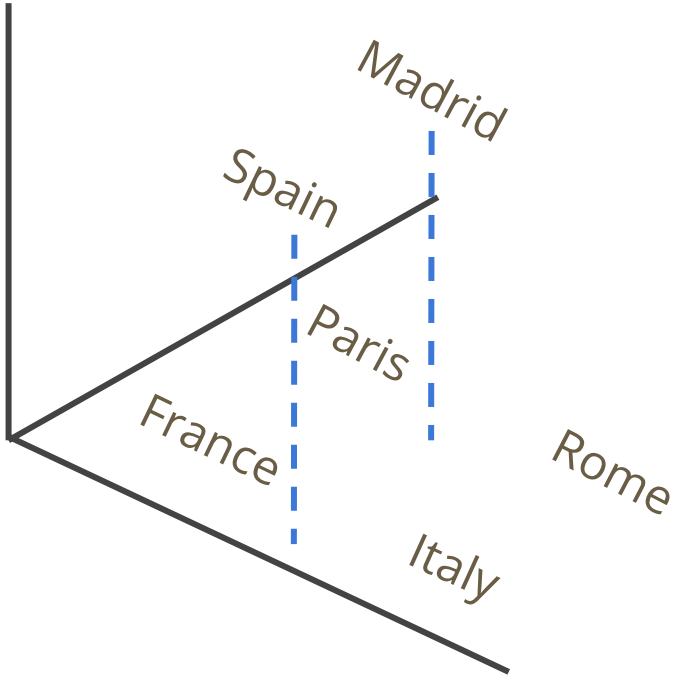
France Italy

What if we now add Madrid and Spain?

Warm Up: working with text data



Warm Up: working with text data



Warm Up: working with text data

Words -> vectors

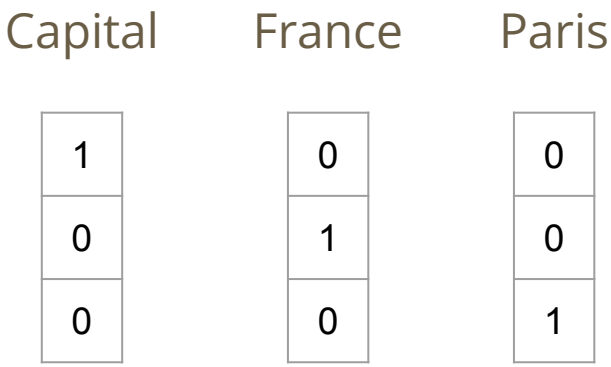
How can we represent the following words:

Capital France Paris

Warm Up: working with text data

Capital France Paris

Warm Up: working with text data



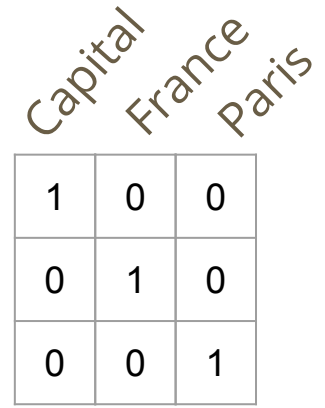
One-hot encoding

Warm Up: working with text data

Capital • France = 0

Paris • France = 0

Capital • Paris = 0



One-hot encoding

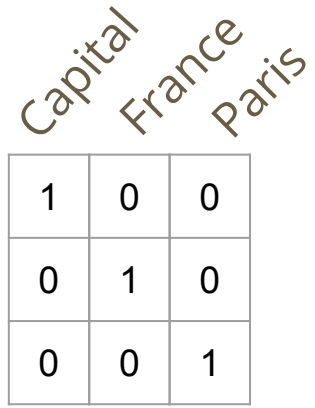
Warm Up: working with text data

But... The capital of France is Paris.

Capital • France = 0

Paris • France = 0

Capital • Paris = 0



One-hot encoding

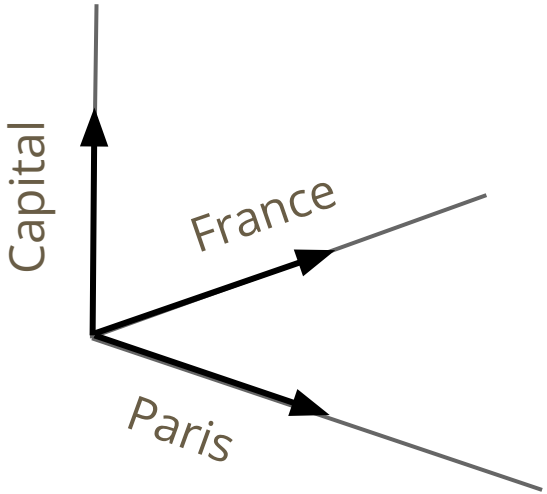
Warm Up: working with text data

But... The capital of France is Paris.

Capital • France = 0

Paris • France = 0

Capital • Paris = 0



Capital	France	Paris
1	0	0
0	1	0
0	0	1

One-hot encoding

Warm Up: working with text data

Capital	France	Paris
1	0	0
0	1	0
0	0	1

One-hot encoding

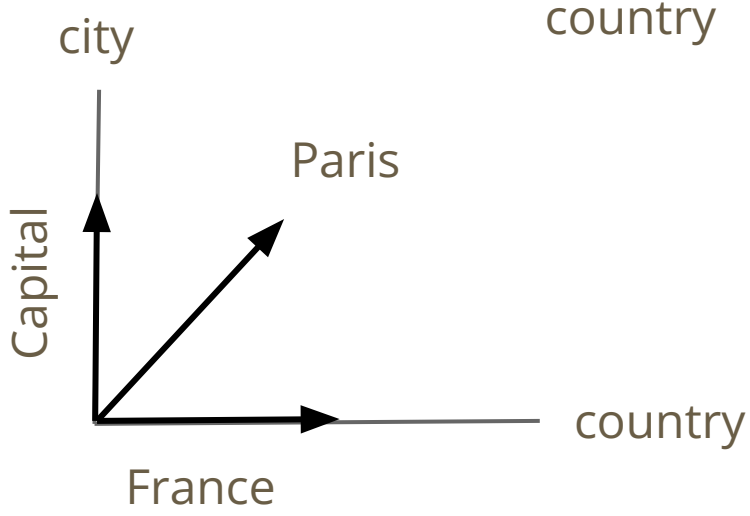
	Capital	France	Paris
city	1	0	1
country	0	1	1

Embedding

Warm Up: working with text data

	Capital	France	Paris
city	1	0	1
country	0	1	1

Embedding



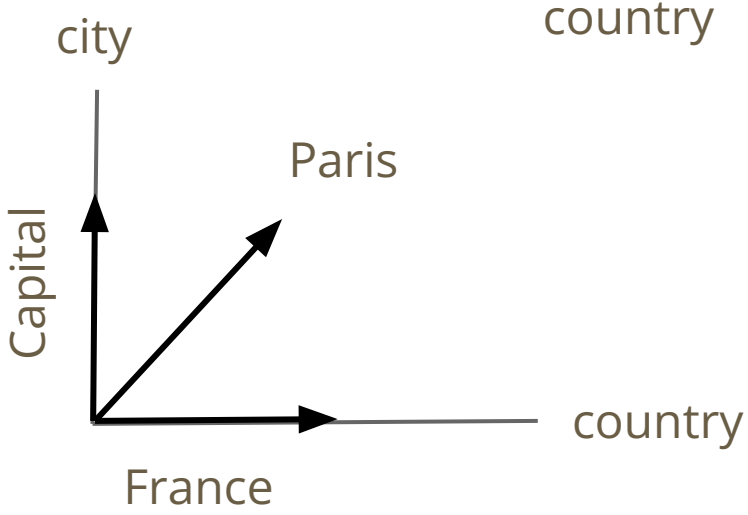
Warm Up: working with text data

Capital + France = Paris

Capital • France = 0

Paris • France = 1

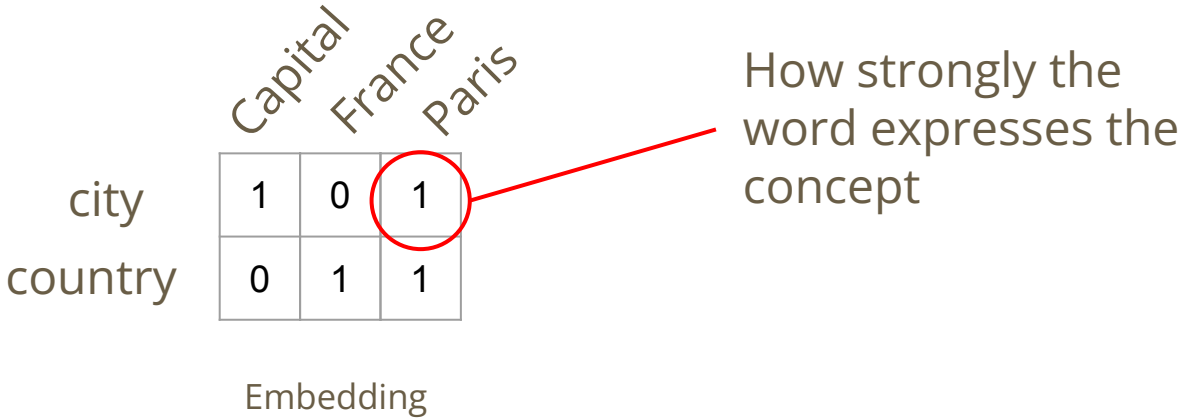
Capital • Paris = 1



	Capital	France	Paris
city	1	0	1
country	0	1	1

Embedding

Warm Up: working with text data



Warm Up: working with text data

	Capital	France	Paris
city	.8	.1	1
country	.1	1	.7

Embedding

Warm Up: working with text data

Capital + France = [.9, 1.1]

Capital • France = .18

Paris • France = .8

Capital • Paris = 0.87

	Capital	France	Paris
city	.8	.1	1
country	.1	1	.7

Embedding

How do we find this embedding space?

Guiding principle: words that appear together a lot should share semantic meaning.

- the **capital** of **france** is **paris**
- **paris** is a beautiful **city**
- **france** is a **country** in europe

	capital	france	paris	city	country
the **capital** of **france** is **paris**	1	1	1	0	0
paris is a beautiful **city**	0	0	1	1	0
france is a **country** in europe	0	1	0	0	1

What if documents are different mixtures of shared underlying patterns?

	capital	**france**	**paris**	**city**	**country**
the **capital** of **france** is **paris**	1	1	1	0	0
paris is a beautiful **city**	0	0	1	1	0
france is a **country** in europe	0	1	0	0	1

$$X \approx Z \times B$$

Docs x words | Docs x patterns | patterns x words

What’s the smallest set of patterns that can accurately approximate X?

$$X \approx Z \times B$$

Docs x words | Docs x patterns | patterns x words

What's the smallest set of patterns that can accurately approximate X?

$$X \approx Z \times B$$

Docs x words | Docs x patterns | patterns x words

If the number of patterns is small we're forced to find reusable structure across documents

capital	france	paris	city	country

capital	france	paris	city	country

1	1	1	0	0
0	0	1	1	0
0	1	0	0	1

=

.8	0	-.6
.4	.7	.6
.4	-.7	.6

X

2	0	0
0	1.4	0
0	0	1

X

.4	.6	.6	.2	.2
0	-5	.5	.5	-.5
-.6	0	0	.6	.6

capital	france	paris	city	country
1	1	1	0	0
0	0	1	1	0
0	1	0	0	1

=

.8	0	-.6
.4	.7	.6
.4	-.7	.6

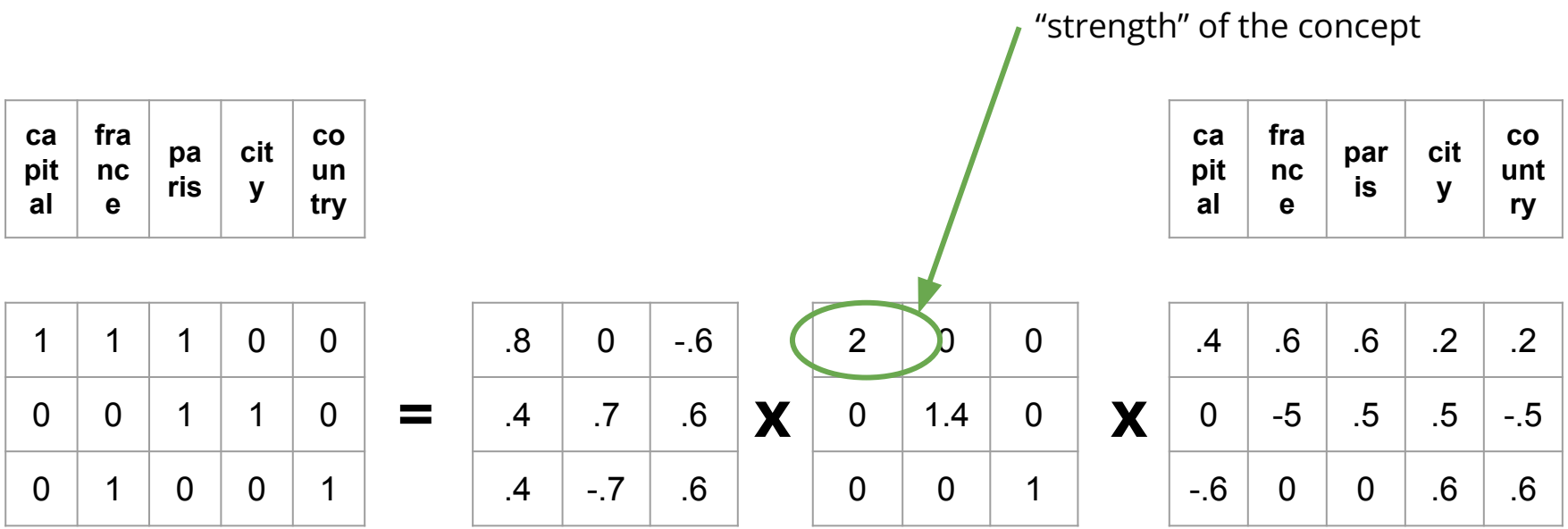
doc-to-concept similarity

x

2	0	0
0	1.4	0
0	0	1

x

capital	france	paris	city	country
.4	.6	.6	.2	.2
0	-5	.5	.5	-.5
-.6	0	0	.6	.6


“strength” of the concept
capital
france
paris
city
country
1 1 1 0 0
0 0 1 1 0
0 1 0 0 1
=
.8 0 -.6
.4 .7 .6
.4 -.7 .6
X
2 0 0
0 1.4 0
0 0 1
X
.4 .6 .6 .2 .2
0 -5 .5 .5 -.5
-.6 0 0 .6 .6

capital	france	paris	city	country
1	1	1	0	0
0	0	1	1	0
0	1	0	0	1

=

.8	0	-.6
.4	.7	.6
.4	-.7	.6

X

2	0	0
0	1.4	0
0	0	1

X

term-to-concept similarity

capital	france	paris	city	country
.4	.6	.6	.2	.2
0	-5	.5	.5	-.5
-.6	0	0	.6	.6

embedding

.8	0	-.6
.4	.7	.6
.4	-.7	.6

X

2	0	0
0	1.4	0
0	0	1

=

Document	Pattern 1	Pattern 2	Pattern 3
capital france paris	1.6330	0	−0.5774
paris city	0.8165	1.0000	0.5774
france country	0.8165	−1.0000	0.5774

Latent Semantic Analysis

	data	information	retrieval	brain	lung
CS-paper-1	1	1	1	0	0
CS-paper-2	2	2	2	0	0
CS-paper-3	1	1	1	0	0
CS-paper-4	5	5	5	0	0
Med-paper-1	0	0	0	2	2
Med-paper-2	0	0	0	3	3
Med-paper-3	0	0	0	1	1

Latent Semantic Analysis

1	1	1	0	0
2	2	2	0	0
1	1	1	0	0
5	5	5	0	0
0	0	0	2	2
0	0	0	3	3
0	0	0	1	1

=

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

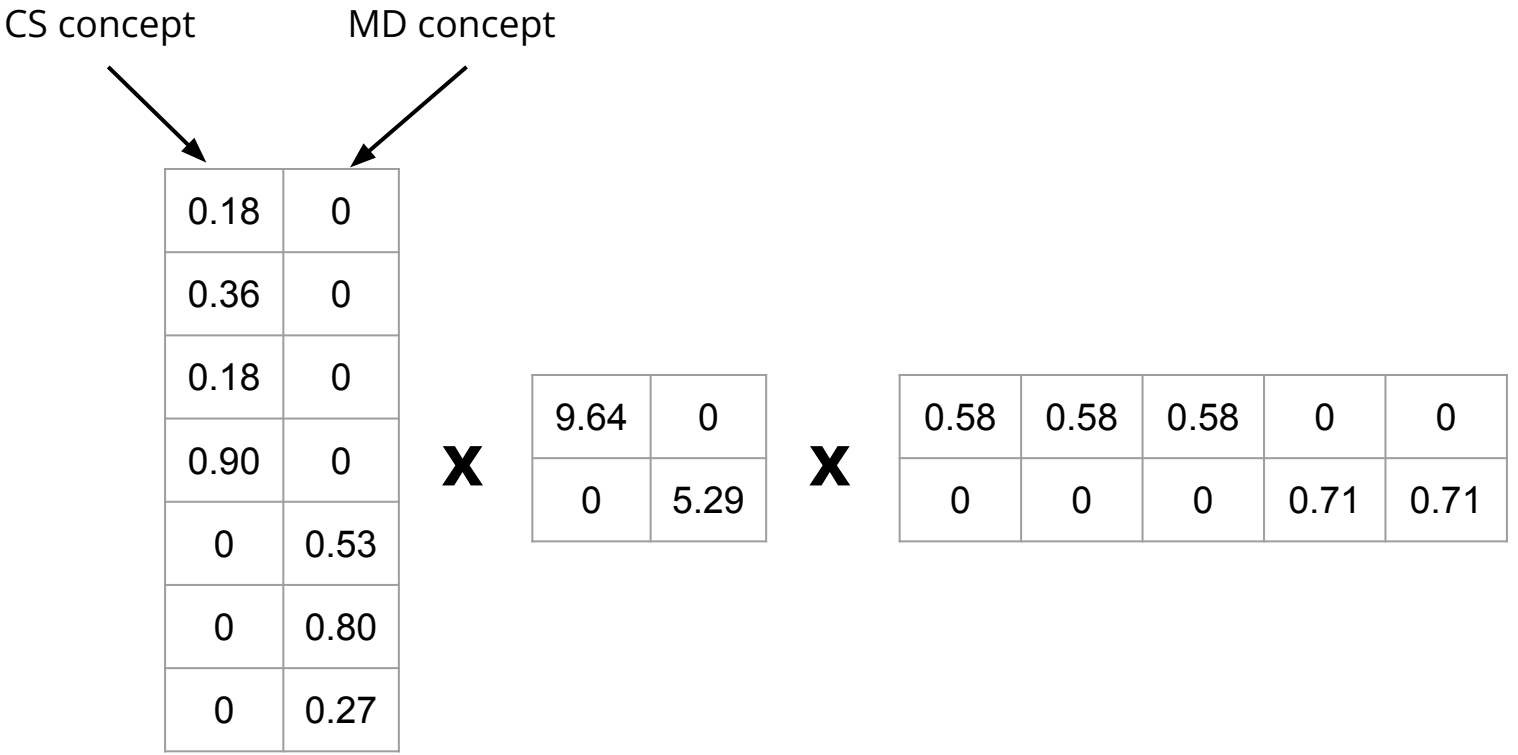
Latent Semantic Analysis

1	1	1	0	0
2	2	2	0	0
1	1	1	0	0
5	5	5	0	0
0	0	0	2	2
0	0	0	3	3
0	0	0	1	1

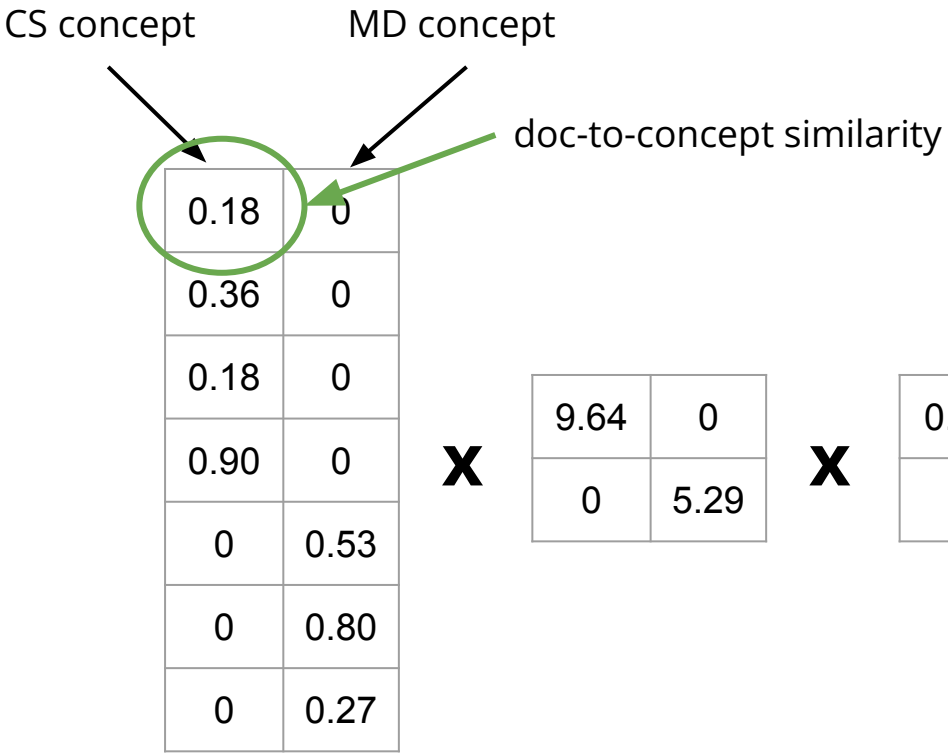


Doc to term similarity

Latent Semantic Analysis



Latent Semantic Analysis



0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
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Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

“strength” of the CS concept

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

"strength" of the each concept

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

"strength" of the each concept

9.64	0
0	5.29

x

term-to-concept similarity

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

"strength" of the each concept

9.64	0
0	5.29

x

term-to-concept similarity matrix

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

What affects the quality of the embeddings found?

Recall the guiding principle: words that appear together should share meaning.

Latent Semantic Analysis

We can better represent each document by:

- The presence of each word (0 / 1)

Latent Semantic Analysis

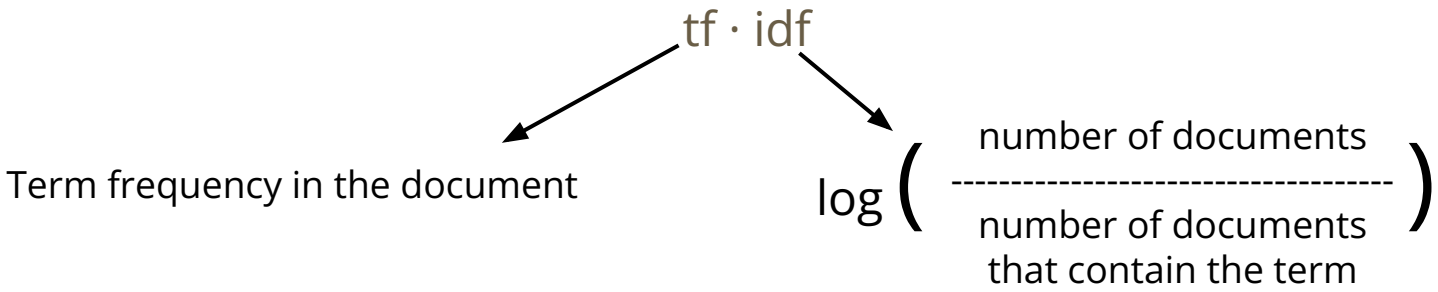
We can better represent each document by:

- The presence of each word (0 / 1)
- Count of the word (0, 1, ...)

Latent Semantic Analysis

We can better represent each document by:

- The presence of each word (0 / 1)
- Count of the word (0, 1, ...)
- Frequency of the word (n_i / Σn_i)
- TfiDf



Exercise: apply tfidf to this matrix

	capital	france	paris	city	country
the **capital** of **france** is **paris**	1	1	1	0	0
paris is a beautiful **city**	0	0	1	1	0
france is a **country** in europe	0	1	0	0	1

TF–IDF weakens the words connecting these documents and strengthens the words distinguishing them. What happens to the latent features / patterns?

Limitations

Concepts are orthogonal...

Concept A · Concept B = 0

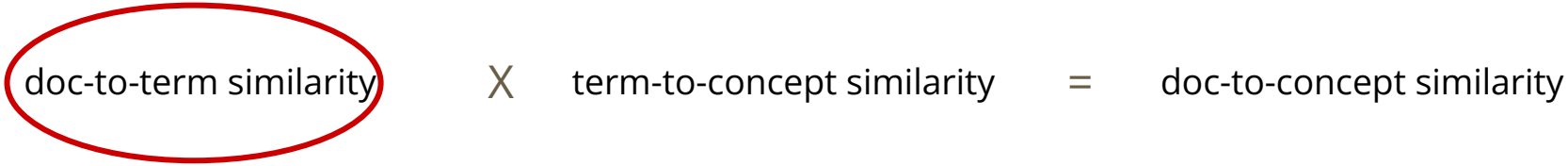
Latent Semantic Analysis

	data	information	retrieval	brain	lung
CS-paper-1	1	1	1	0	0
CS-paper-2	2	2	2	0	0
CS-paper-3	1	1	1	0	0
CS-paper-4	5	5	5	0	0
Med-paper-1	0	0	0	2	2
Med-paper-2	0	0	0	3	3
Med-paper-3	0	0	0	1	1

What concepts best represent these documents? Can we write each document as a vector in a space of concepts?

doc-to-concept similarity

doc-to-term similarity X term-to-concept similarity = doc-to-concept similarity


doc-to-term similarity
X
term-to-concept similarity
=
doc-to-concept similarity

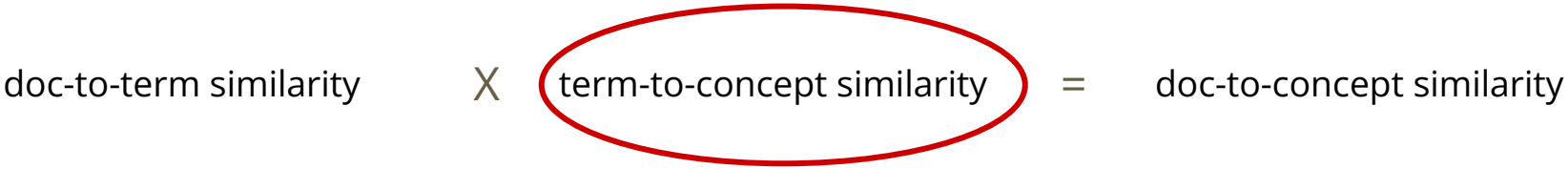
Latent Semantic Analysis

Inputs are documents. Each word is a feature. We can represent each document by:

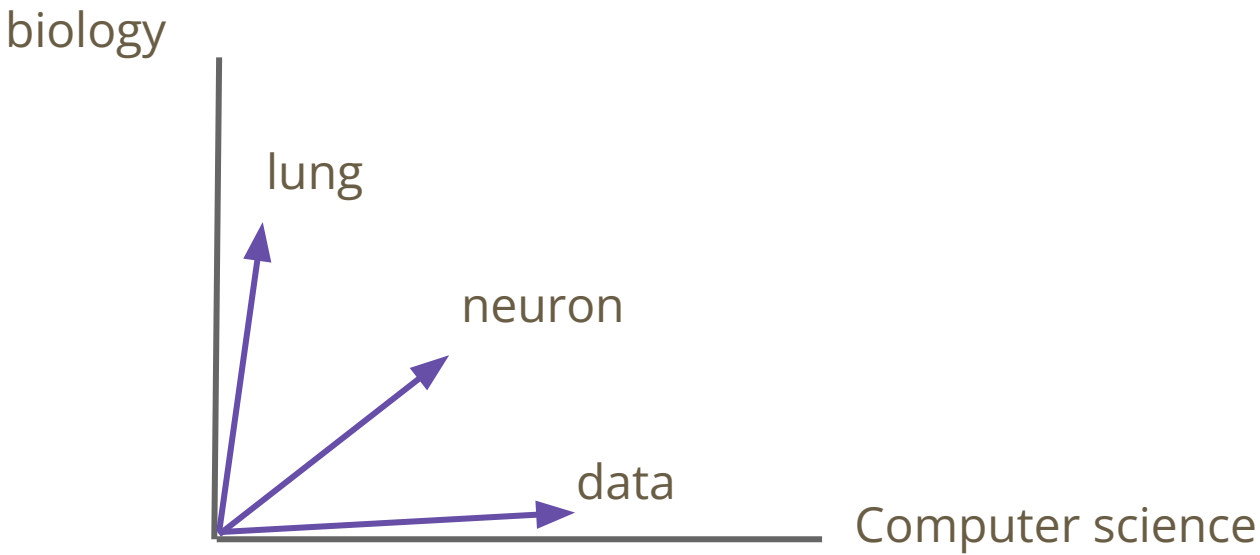
- The presence of each word (0 / 1)

	data	information	retrieval	brain	lung
CS-paper-1	1	1	1	0	0

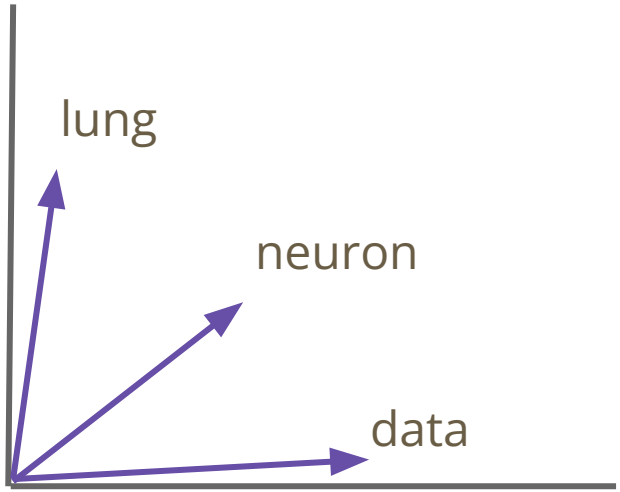
doc-to-term similarity X term-to-concept similarity = doc-to-concept similarity


doc-to-term similarity
X
term-to-concept similarity
=
doc-to-concept similarity

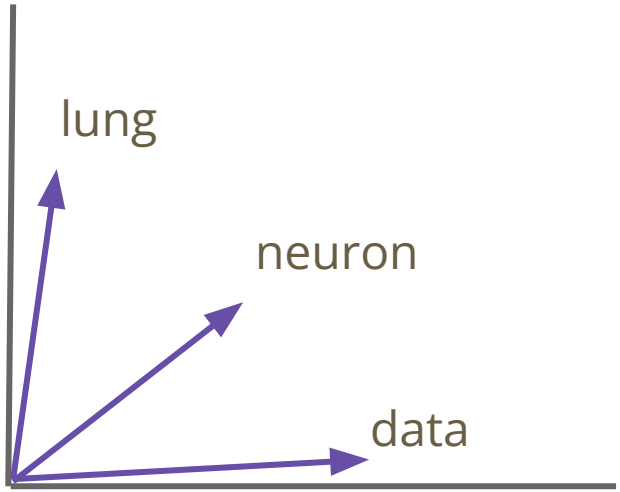
In theory



In practice

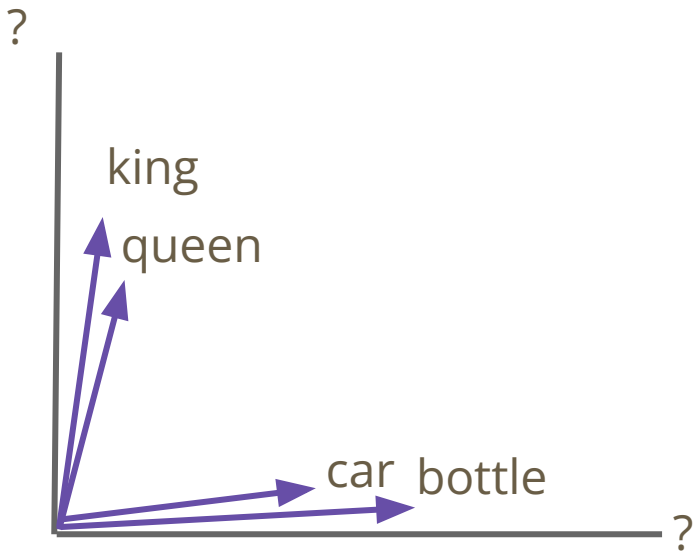


In practice



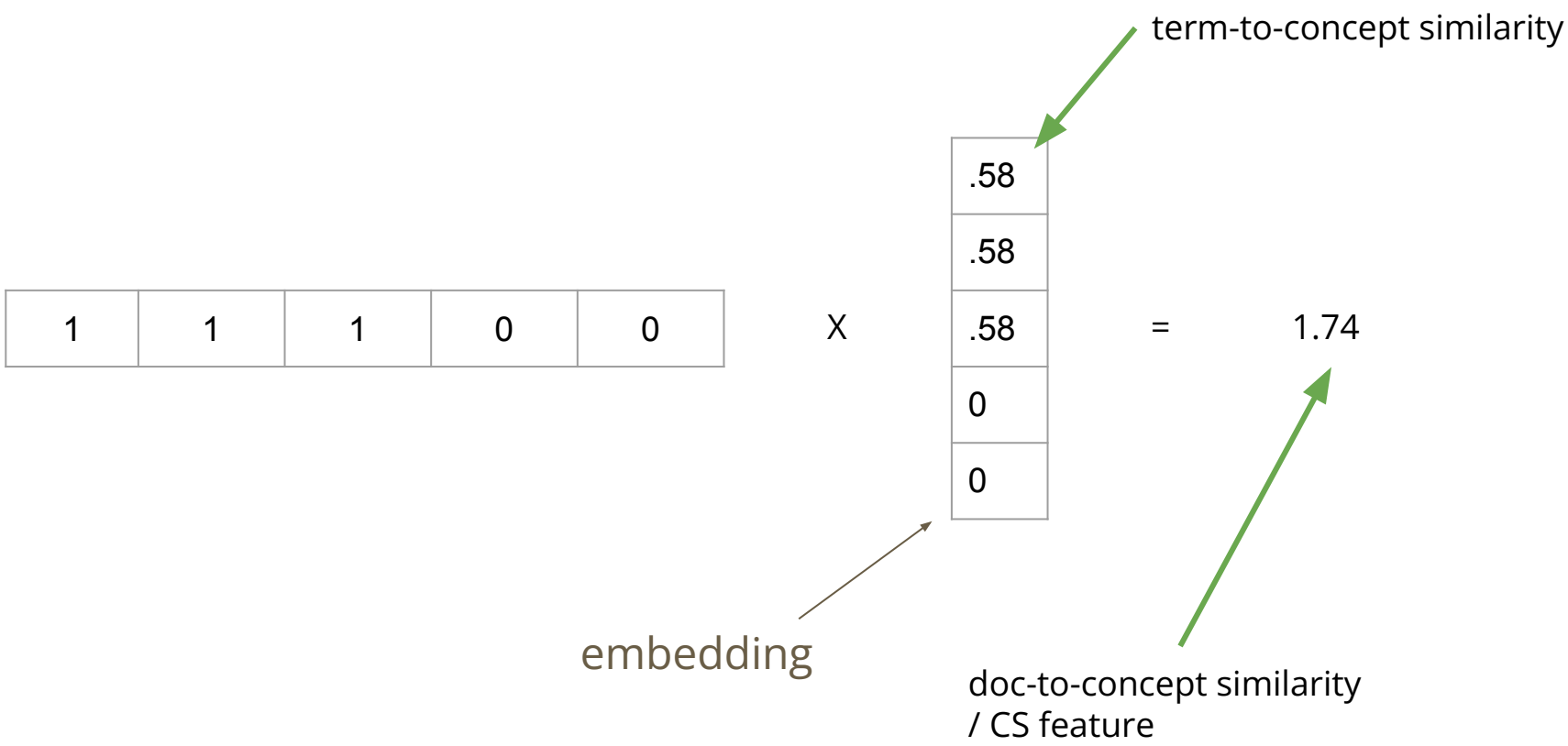
Words with similar semantic meanings should be close

In actual practice



Words with similar semantic meanings should be close

Lots of ways to generate embeddings. SVD is one of them

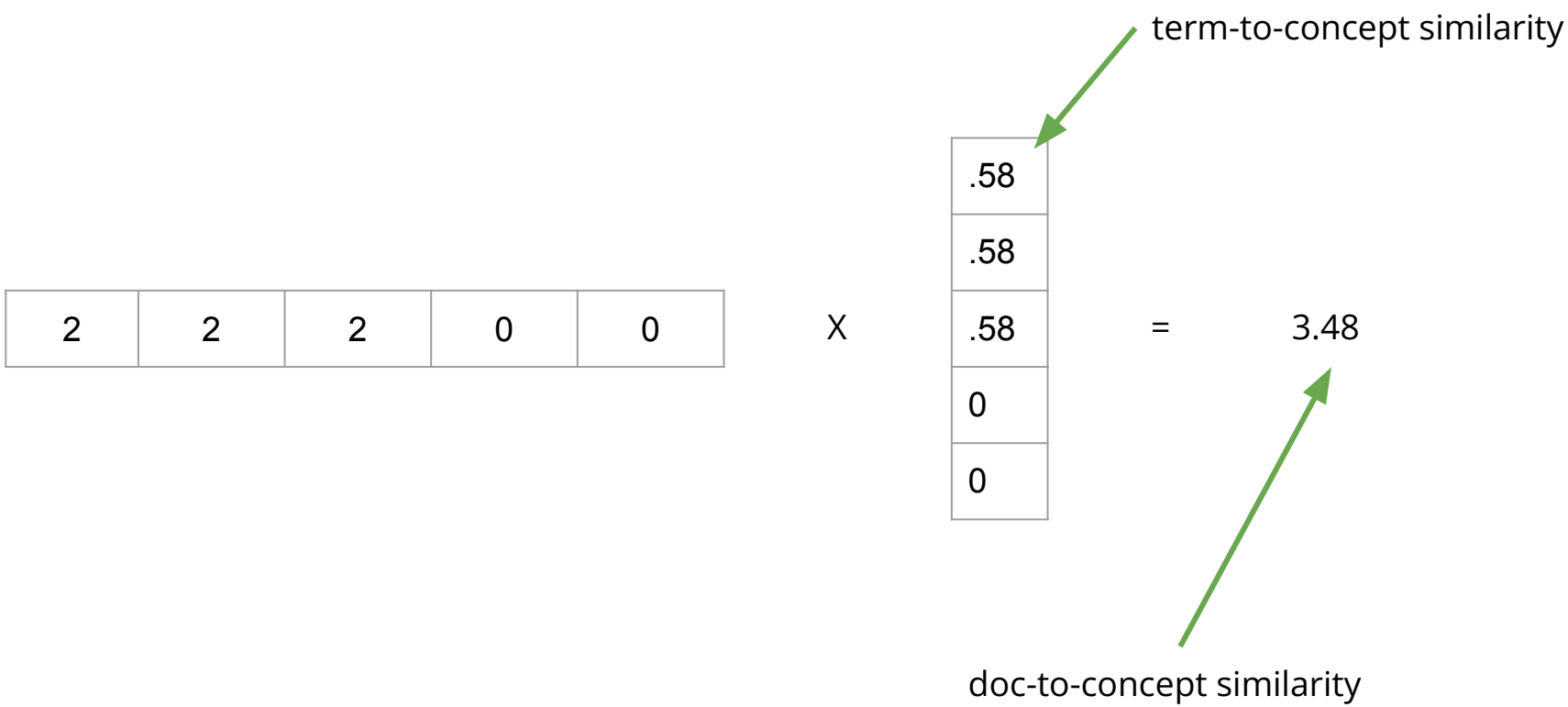

term-to-concept similarity
1
1
1
0
0
X
.58
.58
.58
0
0
=
1.74
embedding
doc-to-concept similarity
/ CS feature

Latent Semantic Analysis

Inputs are documents. Each word is a feature. We can represent each document by:

- The presence of each word (0 / 1)
- Count of the word (0, 1, ...)

	data	information	retrieval	brain	lung
CS-paper-1	2	2	2	0	0


term-to-concept similarity
2
2
2
0
0
X
.58
.58
.58
0
0
=
3.48
doc-to-concept similarity

Latent Semantic Analysis

	data	information	retrieval	brain	lung
CS-paper-1	1	1	1	0	0
CS-paper-2	2	2	2	0	0
CS-paper-3	1	1	1	0	0
CS-paper-4	5	5	5	0	0
Med-paper-1	0	0	0	2	2
Med-paper-2	0	0	0	3	3
Med-paper-3	0	0	0	1	1

Latent Semantic Analysis

1	1	1	0	0
2	2	2	0	0
1	1	1	0	0
5	5	5	0	0
0	0	0	2	2
0	0	0	3	3
0	0	0	1	1

=

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

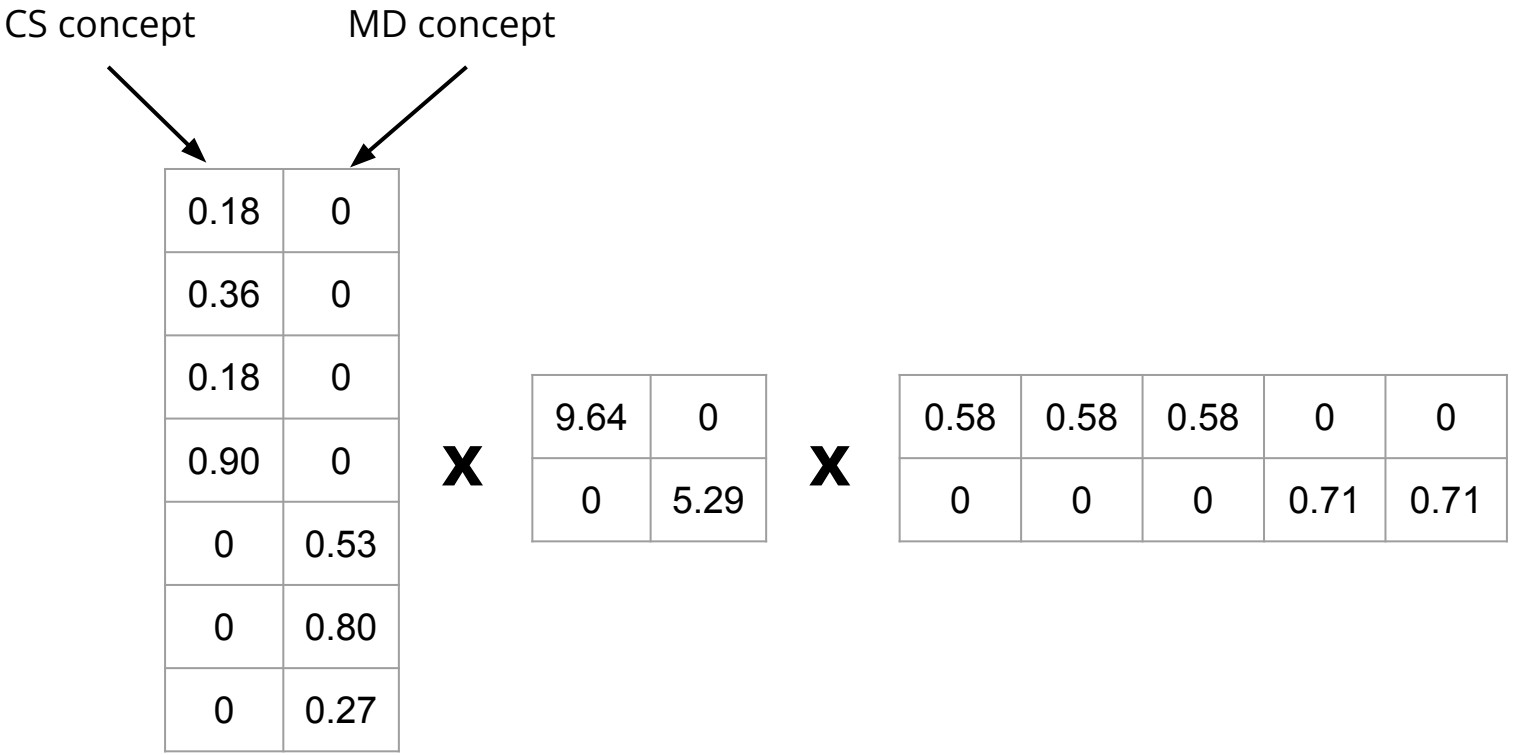
Latent Semantic Analysis

1	1	1	0	0
2	2	2	0	0
1	1	1	0	0
5	5	5	0	0
0	0	0	2	2
0	0	0	3	3
0	0	0	1	1

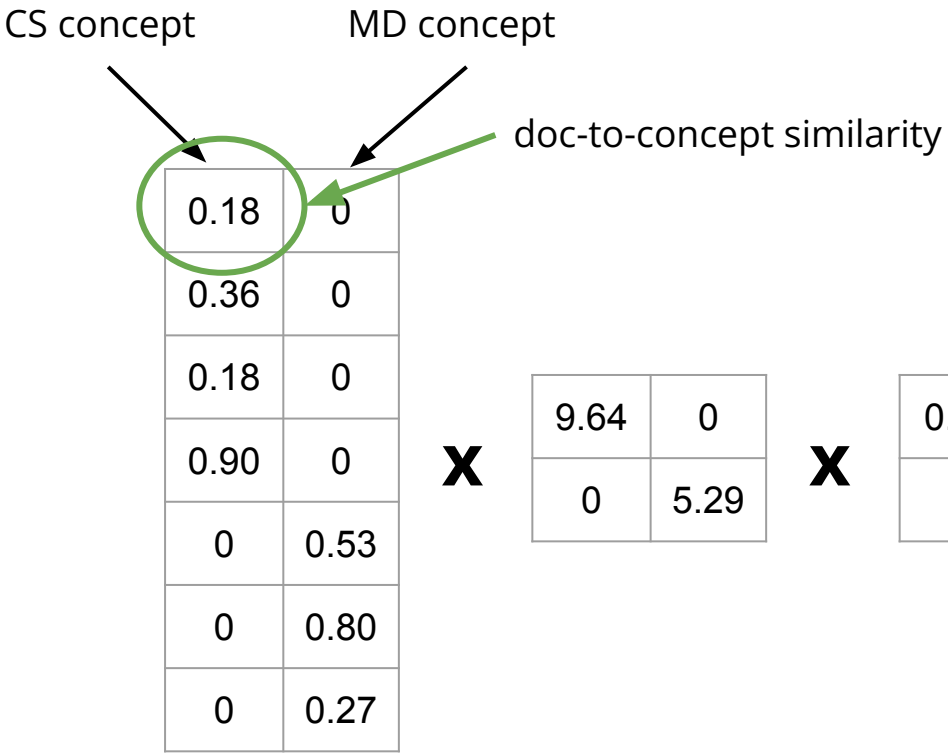


Doc to term similarity

Latent Semantic Analysis



Latent Semantic Analysis



0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

9.64	0
0	5.29

“strength” of the CS concept

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

"strength" of the each concept

9.64	0
0	5.29

x

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

"strength" of the each concept

9.64	0
0	5.29

x

term-to-concept similarity

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

doc-to-concept similarity matrix

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

x

"strength" of the each concept

9.64	0
0	5.29

x

term-to-concept similarity matrix

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

Latent Semantic Analysis

We can better represent each document by:

- Frequency of the word (n_i / Σn_i)
- TfiDf

